

SpaceShare: Leveraging Multimodal Context for Fluid Sharing of Spaces in Video Meetings

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Figure 1: (a) SpaceShare reconstructs the environment in real time, allowing users to independently explore the space, understand each other's location and viewpoint, and communicate attention using virtual pointers. (b) While in use, the system generates multimodal summaries of the space by combining conversation transcripts and contextual cues, enabling users to retrieve specific objects or locations using natural language queries.

Abstract

In video meetings, people not only talk but also share objects and their environments. However, conventional video calls offer limited support for spatial sharing, typically relying on a live video feed. We present SpaceShare, a system that integrates real-time 3D space reconstruction into video meetings and leverages multimodal context to fluidly support shared spatial understanding. The reconstructed 3D environment enables users to independently explore the space, while avatar visualization and shared views convey each participant's location and viewpoints. SpaceShare also stores conversation content and contextual metadata, such as where the conversation took place. This allows users to retrieve spatial features using both physical attributes and conversational context.

CCS Concepts

• **Human-centered computing** → **Collaborative and social computing systems and tools.**

Keywords

Video meetings, remote collaboration, 3D reconstruction, space sharing, shared task space

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1 Introduction

Video meetings are now widely used across many areas. Beyond simple conversation, people increasingly share physical environments through video meetings. For example, they take virtual guided tours [6], view real estate remotely [4], and collaborate on spatial projects [1]. While modern mobile devices can capture scenes in high fidelity, effectively communicating spatial information remains challenging. The video feed is momentary and shows only what is currently in view. Referencing parts of the space is difficult and inefficient, especially when the location is out of view or far from the users. As the shared space becomes larger, problems become more pronounced.

To effectively support space sharing in video meetings, prior work has identified several key aspects: enabling view independence, fostering spatial awareness, and supporting referencing of past scenes. View independence allows users to explore remote environments freely, improving task efficiency and confidence [15], though solutions like 360° video remain limited by fixed viewpoints [1, 16, 17]. Real-time 3D reconstruction using Simultaneous localization and mapping (SLAM) methods [14] or neural techniques like NeRF and Gaussian Splatting [7, 12] may address this by allowing interactive, live exploration. For spatial awareness, systems have used avatars, gaze, and gestures to convey partner context [2, 10], but these become less effective in large spaces. Finally, referencing

previously visited scenes remains challenging. Landmarks and min-imaps help [18], but are often insufficient without integration of conversational context [9].

To address these challenges, we developed SpaceShare, a system that integrates real-time 3D space reconstruction with multimodal contextual features to enhance shared spatial understanding in video meetings. Using SLAM, SpaceShare reconstructs the environment on the fly, enabling both local and remote users to independently explore the shared space. The system visualizes each user's position and viewpoint through virtual avatars and ray-based pointers to facilitate view communication. As users converse, SpaceShare captures not only the 3D environment but also multimodal contextual data, including user locations, and conversation content. It generates panoramic images by stitching keyframes, labels objects in the scene, and stores these representations alongside the conversation history. This enables rich, context-based retrieval of prior scenes. For instance, instead of navigating manually or issuing a generic search like "chair," a user could search for "the chair that reminded us of our childhood memories," leveraging the conversational context to find a specific moment or location.

Our key contribution is the introduction of a novel approach that incorporates multimodal contextual summaries to support spatial sharing and retrieval in remote meetings.

2 SpaceShare Implementation

SpaceShare system consists of four key components: (1) real-time 3D reconstruction, (2) shared spatial meeting interface, (3) multimodal space summarization, and (4) context-based space retrieval.

Real-time 3D Reconstruction. SpaceShare uses the Spectacular AI Visual-Inertial SLAM SDK [13] to reconstruct the environment in real time. This uses both visual and inertial sensor data from the depth camera to generate keyframes, point clouds, meshes, and accurate camera poses representing the local user's environment from a first-person perspective. To ensure a coherent spatial representation, all geometry is aligned within a unified world coordinate system. Remote users do not perform local reconstruction but instead access a shared model hosted on the server through Unity's Mirror networking framework.

Shared Spatial Meeting Interface. The user interface for the meeting system is built with considerations for supporting view awareness and referencing (See Fig. 1). Virtual avatars represent each participant's position and orientation. Ray-based pointers visualize each user's current view direction, enabling mutual view awareness. Users can also leave anchors or marks in the space to draw attention to specific regions.

Multimodal Space Summarization. SpaceShare summarizes key spatial moments to support efficient recall and navigation. During the SLAM process, the system generates keyframes based on visual differences between frames. These keyframes are then grouped according to spatial proximity and temporal data. For each group, a panoramic image is created using OpenCV's stitching tools [11], leveraging the associated position and orientation metadata. To enrich these visual summaries, the system uses YOLO-World [5] to detect and label objects and OpenAI GPT-4V [8] to generate descriptive captions for each panorama, which are stored alongside their corresponding spatial metadata. In parallel, SpaceShare uses

the OpenAI Speech-to-Text API to transcribe the meeting conversation. The transcript is timestamped and aligned with the keyframe and panorama data, creating a rich, multimodal representation of the session. These multimodal space summaries are compiled into a searchable memory database.

Context-based Space Retrieval. SpaceShare allows users to search for previously discussed locations using natural language queries. The system leverages the meeting transcript and spatial index to match semantic phrases (e.g., "the chair we joked about") to specific panoramic views and 3D locations. The system uses OpenAI GPT-4o to interpret user queries and identify relevant objects along with their context—such as their position in the reconstructed space and the associated panoramic image.

System Setup We implemented the portable system using a tablet PC (ASUS ROG Flow Z13, AMD Ryzen 9 7940HS CPU, NVIDIA RTX 4070 GPU, 16 GB RAM) paired with an Intel RealSense D455 depth camera for SLAM reconstruction. This setup resembles consumer tablets equipped with depth cameras, such as the iPad Pro. To ensure smooth performance, the computation for 3D reconstruction was offloaded to a desktop PC (Intel Core i9-9900K CPU, NVIDIA RTX 2080 GPU, 32 GB RAM).

3 Preliminary User Study

To gain early insights into how our system supports fluid spatial sharing and contextual recall, we conducted an exploratory user study with 16 participants (10 male, 6 female). Participants were paired in groups of two, with one acting as the local user who scanned the environment, and the other as the remote user.

The participants completed two tasks with and without using the context-based space retrieval feature. Half of the users started without using the space retrieval feature and the other started with the feature. In the first task, participants explored the environment together in real time. The local user described the space and nearby artworks while moving through the scene, and the remote user received both the live video feed and a continuously updated 3D reconstruction. They were asked to explore and discuss the artworks and select three of them that they liked the most. In the second task, the participants were asked to locate five previously seen artworks from a provided list within 12 minutes.

After using the system, participants completed the System Usability Scale (SUS) questionnaire [3] and provided feedback on their experiences. The mean SUS scores were 55.47 with the space retrieval feature and 53.91 without it, indicating a marginal level of usability in both cases. In the second task, participants located more artworks when using the context-based space retrieval feature (average: 4.44) than without it (average: 3.62). Additionally, ten participants explicitly noted the usefulness of the retrieval feature.

4 Discussion and Future Work

While our preliminary study demonstrated the potential of SpaceShare for enhancing spatial recall and collaboration in remote meetings, the SUS scores indicate that the overall interface and interaction design still require improvement. Future work will include iterative design refinements to improve usability and a more comprehensive user study to gain deeper insights into the effects of the space summarization and retrieval features.

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